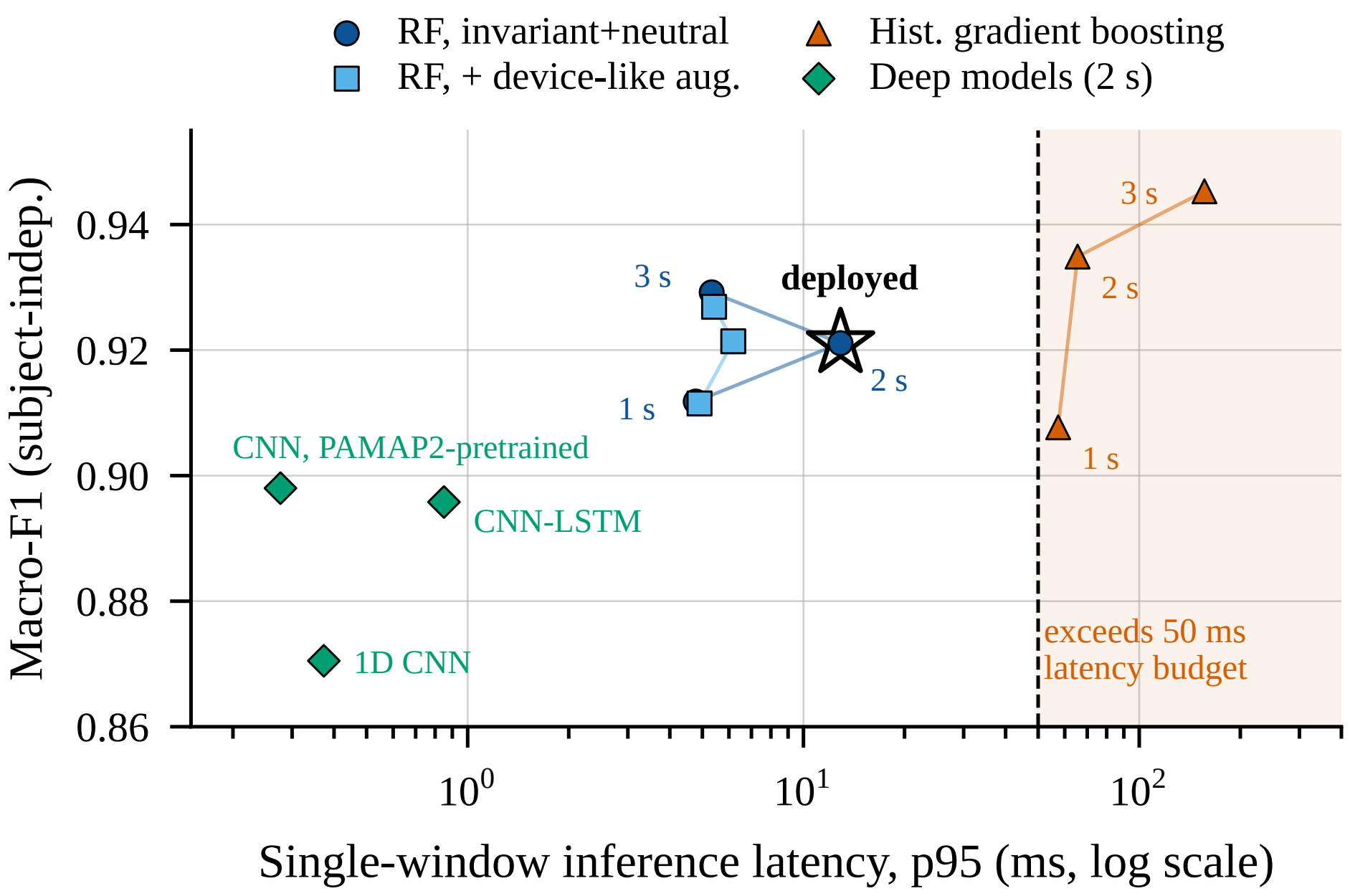
RF, invariant+neutral
RF, + device-like aug.
Hist. gradient boosting
Deep models (2 s)
Macro-F1 (subject-indep.)
0.94
0.92
0.90
0.88
0.86
3 s
deployed
2 s
1 s
CNN, PAMAP2-pretrained
CNN-LSTM
1D CNN
3 s
2 s
1 s
exceeds 50 ms
latency budget
10^0
10^1
10^2
Single-window inference latency, p95 (ms, log scale)